

Seeding MCMC/HMC algorithms

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Abstract

The primary goal of this work is to illustrate the need of seeding MCMC/HMC algorithms when dealing with unobserved/latent variables.

1 Introduction

In real-world scenarios, researchers and practitioners often aim to infer

2 Literature Review

[Maa et al. \(1996\)](#)

3 Methodology

4 Simulations

For a discrete-time setting, let us assume that a population experience an exponential growth with time-varying rate r_t , that is, $N_{t+1} = e^{r_t} N_t$ for $t = 1, 2, \dots, T - 1$. We can rewrite the time-varying rate as $r_t = f_t - m_t$, where $f_t, m_t \in \mathbb{R}^+$ are the fertility and mortality rates at time t , respectively. Now, at each time t , we can think of abundance before and after mortality, say N_t^{pre} and N_t^{post} , respectively. Thus, if M_t denotes the total mortality we have that $N_t^{post} = N_t^{pre} - M_t$.

Given the aforementioned population demographics, we can propose the following stochastic population mechanism:

$$\begin{aligned} M_t &\sim \text{Bin} \left(N_t^{pre}, 1 - e^{-m_t} \right) \\ N_{t+1}^{pre} &\sim \text{Poi} \left(e^{f_t} N_t^{post} \right) \end{aligned}$$

From a Bayesian perspective

$$\begin{aligned} \log(N_1^{pre}) &\sim \text{Poi} \left(e^{LN_0} \right) \\ \log(a_t) &\sim \text{N} \left(\mu_{la}, \sigma_{la} \right) \\ \log(b_t) &\sim \text{N} \left(\mu_{lb}, \sigma_{lb} \right) \\ \log(f_t) &\sim \text{N} \left(\mu_{lf}, \sigma_{lf} \right) \end{aligned}$$

with

$$\begin{aligned} \mu_{lm} &\sim \text{N} \left(m_{lm}, sd_{lm} \right) \\ \sigma_{lm} &\sim \text{N} \left(0, sd_{lm} \right) \\ \mu_{lf} &\sim \text{N} \left(m_{lf}, sd_{lf} \right) \\ \sigma_{lf} &\sim \text{N} \left(0, sd_{lf} \right) \end{aligned}$$

5 Concluding Remarks

References

- Maa, J.-F., Pearl, D. K., and Bartoszyński, R. (1996). Reducing multi-dimensional two-sample data to one-dimensional interpoint comparisons. *The Annals of Statistics*, 24(3):1069–1074.